Automatic Question Answer Generation Based on Gujarati Paragraph

PhD Weekly Progress Presentation — 3

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February 2026

Work Completed

- Completed Gujarati Question Answer dataset creation
- Structured dataset in JSON format
- Performed preprocessing and validation
- Conducted dataset statistical analysis
- Applied Bloom's Taxonomy based Question Type Classification

Gujarati QA Dataset Overview

- Dataset created from Gujarati paragraph-based educational content
- Domains covered:
  - History
  - Literature
  - Wildlife
  - Religion
- Designed for Extractive Question Answering
- Suitable for Small Language Models (SLMs)

Dataset Statistics

Metric	Value
Total Context Paragraphs	7
Total Question–Answer Pairs	190
Average QA per Context	27
Language	Gujarati
Format	JSON
Answer Type	Extractive Span

Difficulty Distribution

- Easy: 70 questions (36.8%)
- Medium: 61 questions (32.1%)
- Hard: 59 questions (31.1%)

Dataset Example

Context

અમદાવાદ ગુજરાત રાજ્યનું સૌથી મોટું શહેર છે. તેની સ્થાપના સુલતાન અહમદ શાહે કરી હતી.

Question

અમદાવાદની સ્થાપના કોણે કરી હતી?

Answer

સુલતાન અહમદ શાહે

Bloom's Taxonomy Based Classification

- Applied Statistical NLP model for Question Type Prediction
- Feature Extraction: TF-IDF
- Classifier: Multinomial Logistic Regression
- Evaluation: 10-Fold Cross Validation

Cross Validation Accuracy

Language	CV Accuracy
Gujarati	99.55%
Hindi	95.33%
English	100%

Why Statistical Validation is Important?

- Ensures balanced cognitive distribution
- Validates dataset quality quantitatively
- Confirms Bloom-level categorization reliability
- Strengthens research foundation before QA model training

Research Alignment Flow

Dataset Creation

$\downarrow$

Question Type Classification (Statistical Validation)

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Baseline Extractive QA Model Training

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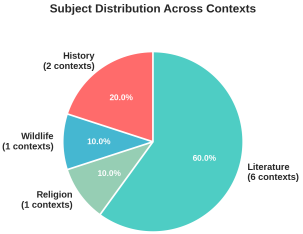
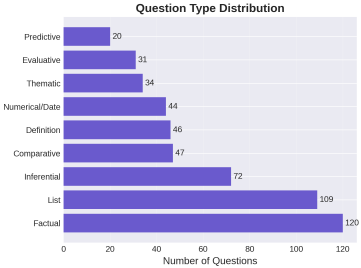
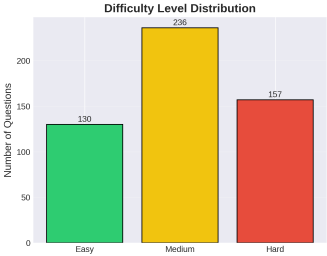
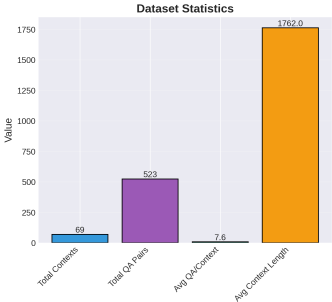
Future: Automatic Question Generation + Cognitive Tagging

Dataset Analysis Visualization









Work Plan for Next Week

- Final dataset refinement after statistical validation
- Prepare dataset for model training
  - Tokenization
  - Train–Validation split
  - Input formatting
- Train baseline Extractive QA Model
  - Transformer-based architecture
  - Hyperparameter tuning
- Analyze performance across difficulty levels

Thank You

Questions?

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